Safety systems · Japan

JR東日本

Japanese signals give a **speed class** for the section ahead. On the TSW **Tadami Line** (single track, 65 km/h) the main signals have three aspects, **Clear**, **Caution** (45 km/h) and **Stop**, with a four-lamp **distant signal** before each station and **repeaters** where the view is short. The **JR East KiHa 40** runs under **ATS-SN** and the **EB** vigilance device. Busier JR lines add **Reduce speed** and **Restricted speed** on five-lamp heads (shown for reference). Quadrant controls are from TheJAG's KiHa 40 profile; "gear lever" means the Quadrant's gear lever (the KiHa's own gear is on the flap lever).

Safety systems

ATS-SN

自動列車停止装置 S type, N variant

A coil about 600 m before a signal at stop rings the **bell**: brake and press **ATS Acknowledge** (確認扱い) within **5 s**, or the emergency brake applies. The **alert sound** then continues. A coil **at** the signal brakes at once if it is passed at stop, and a departure coil does the same if you start against a starting signal at stop.

WHERE The whole Tadami Line in TSW

QUADRANT **Gear lever** = ATS Acknowledge; **left aux** = ATS and EB power; **round aux** = Alert Release (警報持続解除)

ATS-P / Ps

Pattern-type ATS

Checks a braking curve to signals and restrictions and brakes automatically; no acknowledge. On the real line Ps covers the Aizu-Wakamatsu and Koide station areas; **TSW simulates SN only**.

WHERE P: Tokyo area and main lines. Ps: JR East regional lines (Sendai, Niigata areas)

EB

EB装置 (Emergency Brake device)

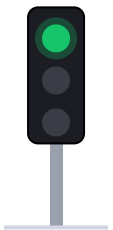
Above **15 km/h**, with no throttle, brake or horn movement for about **60 s**, a buzzer and lamp: reset within about **5 s** or the emergency brake applies.

WHERE Most JR trains, the KiHa 40 included

QUADRANT **Left throttle button** = EB Reset

Tadami Line signals

Main signals (home 場内, starting 出発, block 閉塞): three lamps. Distant signal (遠方信号機): square backboard, four positions, one left empty.



Clear (進行)

Line speed.

DRIVE Drive at the line speed.



Caution (注意)

Next signal at stop: 45 km/h.

DRIVE Brake to 45; prepare to stop.



Stop (停止)

Do not pass.

DRIVE Stop before the signal.



Distant: Clear

The home signal shows Clear.

DRIVE Keep going.



Distant: Advanced caution

The home signal shows Caution.

DRIVE Prepare for 45 at the home signal.



Distant: Caution

The home signal is at stop: 45 km/h.

DRIVE Brake to 45, prepare to stop.

Repeaters, route indicators and the five-lamp head

Five-lamp heads (yellow, yellow, red, yellow, green) are not on the Tadami Line; typical JR speeds shown.



Repeater: proceed (進行中継)

Vertical: the main signal ahead shows Clear.

DRIVE Keep going.



Repeater: restricted (制限中継)

Diagonal: Caution (or Reduce / Restricted speed).

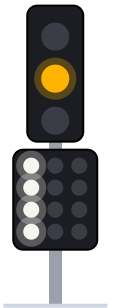
DRIVE Prepare to slow down.



Repeater: stop (停止中継)

Horizontal: the main signal is at stop.

DRIVE Prepare to stop.



Route indicator (進路表示機)

Under the head, lit with the signal: the lit column is the route (left here).

DRIVE Check it is your track.



Reduce speed (減速)

Next signal at caution: about 65 km/h.

DRIVE Brake to the speed at the signal.



Restricted (警戒)

Short overlap to the next stop signal: 25 km/h.

DRIVE Max 25, stop at the next signal.

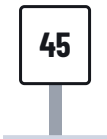
Sources: Dovetail Games: Tadami Line / KiHa 40 driver's manual · MLIT 鉄道に関する技術上の基準を定める省令 · ja.wikipedia: 自動列車停止装置, 日本の鉄道信号, EB装置, 只見線 · en.wikipedia: Japanese railway signals

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Signs on the Tadami Line



Speed limit (速度制限標識)

A curve or points limit from here.

DRIVE Be at the speed by the sign.



Limit end (速度制限解除標識)

The limit ends.

DRIVE Accelerate once the rear has passed.



Departure coil

ATS departure coil: do not pass with the starting signal at stop.

DRIVE Wait for the starting signal.



Whistle sign

Sound the horn.

DRIVE Horn (right throttle button).



Stop position (停止位置目標)

Stop here with a train of this many cars.

DRIVE Stop at your car count.

How to drive under ATS-SN and the EB

ATS-SN: approaching a stop signal

1. About 600 m before a signal at stop the **bell** rings.
2. Brake handle to Lap or Apply, and within 5 s press **ATS Acknowledge** (gear lever).
3. The **alert sound** continues: stop before the signal.
4. Once stopped (or when the signal clears) press **Alert Release** (round aux), then drive on.

ATS emergency brake

1. Passing a signal at stop, or no acknowledge in time: emergency brake.
2. Brake handle to **Emergency**, wait for a complete stop.
3. Press **ATS Acknowledge** (gear lever).
4. Release the brake once the brake pipe has recovered; wait for the signal.

EB (vigilance)

1. Above 15 km/h, moving the throttle, brake or horn resets the timer.
2. After about 60 s without one the EB buzzer sounds: press **EB Reset** (left throttle button).
3. Missed: emergency brake. Brake handle to Emergency, the warning stops; then release.

Before departure

1. **ATS and EB power** on (left aux): the self-test rings the bell for 2 s, then the alert sound.
2. Press **Alert Release** (round aux) to silence it.
3. Check the starting signal and the departure time, close the doors (RUD TRIM), release the brake, notch up.

Single track

- The Tadami Line is single track with crossing loops (Nishi-Wakamatsu, Aizu-Bange, Aizu-Miyashita, Aizu-Kawaguchi).
- At a loop the starting signal governs: wait for it to clear.
- The departure coil brakes the train if you start against it.

Speed

- Line speed 65 km/h; Caution 45 km/h.
- Speed signs set curve and points limits; higher speed only once the rear has passed the end sign.
- ATS-SN does not check your speed: brake to stop yourself.

Remember: ATS-SN only checks that you acknowledged; it does not watch your speed afterwards. The coil at the signal is the last line of defence and stops the train at once.

Sources: Dovetail Games: Tadami Line / KiHa 40 driver's manual · MLIT 鉄道に関する技術上の基準を定める省令 · ja.wikipedia: 自動列車停止装置, 日本の鉄道信号, EB装置, 只見線 · en.wikipedia: Japanese railway signals

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